

Flight Delay Severity Prediction Using Deep Tabular Models and a Calibrated TabTransformer

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Abstract—Flight delay prediction is an important problem in air transportation because delays affect passengers, airlines, airport operations, crew scheduling, and downstream flight planning. This paper presents a deep learning pipeline for multiclass flight delay severity prediction using real-world flight records from 2023, 2024, and 2025 obtained from the U.S. Bureau of Transportation Statistics. The project processes approximately 21 million raw flight records and produces a model-ready dataset of 20,588,160 flights. The final target is defined as a three-class severity problem: On time, Delay, and Long delay.

Three PyTorch deep learning models were compared: an MLP with categorical embeddings, a Wide and Deep neural network, and a TabTransformer-style model with categorical and numerical feature tokens. Because the dataset is imbalanced, Macro F1-score was selected as the main evaluation metric instead of accuracy alone. The final calibrated TabTransformer achieved the best test Macro F1-score of 0.4346 and the best balanced accuracy of 0.4425. Although the raw TabTransformer achieved higher accuracy, the calibrated version improved minority-class detection, especially for Long delay. Finally, a Python graphical inference interface was implemented to load the trained calibrated TabTransformer and estimate calibrated delay severity probabilities for selected flights.

Index Terms—flight delay prediction, deep learning, tabular data, TabTransformer, Wide and Deep learning, class imbalance, Macro F1-score, calibrated probabilities, PyTorch

I. INTRODUCTION AND MOTIVATION

Flight delays are a persistent operational problem in commercial aviation. A delayed flight can affect passenger satisfaction, aircraft utilization, crew scheduling, airport resources, gate assignment, and connecting flights. Even when a delay is not severe, its effect can propagate through later operations and increase uncertainty for airlines and passengers. For this reason, being able to estimate delay risk before the flight takes place can support better planning and more informed operational decisions.

The U.S. Bureau of Transportation Statistics (BTS) commonly considers an arrival delayed when the aircraft arrives 15 minutes or more after the scheduled arrival time [2]. However, a binary delayed/not-delayed classification does not fully describe the severity of the event. A flight delayed by 20 minutes is not equivalent to a flight delayed by more than 90 minutes. The second case is more disruptive and more relevant from an operational perspective.

For this reason, this work models flight delay prediction as a three-class severity classification problem. The final target classes are:

- **On time:** arrival delay less than or equal to 15 minutes.
- **Delay:** arrival delay from 16 to 90 minutes.
- **Long delay:** arrival delay greater than 90 minutes.

The project originally considered a five-class target. However, that definition created severe imbalance, especially for very long delay classes. This made the classification problem unstable and caused early models to collapse toward either the majority class or minority classes depending on the imbalance strategy. The final three-class target was selected to preserve operational meaning while making the task more learnable.

The main challenge in this project is that delay severity is naturally imbalanced. Most flights are on time, while long delays are much less frequent. This makes accuracy an insufficient evaluation metric. A model can obtain high accuracy by predicting mostly the majority class and still fail to detect the minority Long delay class. Therefore, this paper uses Macro F1-score as the main metric because it gives equal importance to each class.

The main contributions of this paper are:

- A complete preprocessing pipeline for a large 2023–2025 flight delay dataset.
- A leakage-safe feature set using only pre-flight or schedule-based information.
- A stratified random split designed to preserve class proportions across train, validation, and test sets.
- A comparison of three deep learning models for tabular delay severity prediction.
- A calibrated TabTransformer model that achieves the best balanced performance.
- A Python graphical interface for estimating calibrated delay probabilities from user-selected flight features.

II. STATE OF THE ART

Flight delay prediction has been studied using statistical learning, machine learning, and deep learning methods. Previous work has shown that aviation delay patterns can be learned from operational data, but prediction quality depends strongly on the information available at prediction time. Features such as previous aircraft delay, weather, air traffic congestion, and airport operations can significantly improve performance, but they may not always be available before a flight is scheduled to depart.

Machine learning approaches for flight delay prediction often use large aviation datasets and classification or regression

models. Big data-driven flight delay studies have shown that machine learning can support delay forecasting and operational planning [3]. However, many highly predictive variables are also close to the actual delay event, which can create data leakage if they are used incorrectly. In this work, leakage prevention is an explicit design constraint: only pre-flight or schedule-based information is used as model input.

Deep learning for tabular data remains a challenging research area. Unlike images or text, tabular datasets contain mixed numerical and categorical variables with heterogeneous meanings. A common strategy is to represent high-cardinality categorical variables with learned embeddings. This is useful in this project because features such as route, origin airport, destination airport, and carrier contain many categories.

Wide and Deep learning was introduced to combine direct linear memorization with deep nonlinear generalization [4]. In this project, the Wide and Deep model is used to combine direct numerical signals with dense categorical embeddings. This allows the model to learn both simple relationships and more complex interactions.

Transformer-based models have also been adapted to tabular data. The Transformer architecture introduced self-attention as a mechanism for learning relationships between tokens [5]. TabTransformer applies this idea to tabular data by transforming categorical embeddings into contextual representations through self-attention [6]. FT-Transformer and related studies have further evaluated Transformer-based architectures for tabular problems and shown that attention-based tabular models can be competitive under appropriate settings [7].

This project builds on these ideas by comparing an embedding-based MLP, a Wide and Deep neural network, and a TabTransformer-style architecture on a large, imbalanced, multiclass flight delay severity task.

III. METHODOLOGY

A. Dataset Description

The dataset was obtained from the Airline On-Time Performance Data database provided by the U.S. Bureau of Transportation Statistics (BTS) through the TranStats platform [1]. This database contains scheduled and actual departure and arrival times reported by certified U.S. air carriers. According to BTS, the database is available from 1987 to 2025 with monthly frequency.

Although historical records are available since 1987, this work uses records from 2023, 2024, and 2025. These years were selected because they provide recent operational patterns while substantially increasing the amount of training data compared with a single-year dataset. In total, 36 monthly CSV files were used.

The raw dataset contains approximately 21 million flight records. After cleaning and feature selection, the final model-ready dataset contains 20,588,160 flights. Table I summarizes the main dataset versions generated during the pipeline.

Cancelled flights were removed because they do not have a valid completed arrival delay value. Rows with missing arrival delay were also removed. Negative arrival delay values were

TABLE I
DATASET VERSIONS GENERATED DURING PREPROCESSING

Dataset	Rows	Description
Raw unified dataset	~21M	36 monthly CSV files
Model-ready dataset	20,588,160	Cleaned and selected features
Train split	14,411,712	70% stratified split
Validation split	3,088,224	15% stratified split
Test split	3,088,224	15% stratified split

converted to zero because they indicate early arrivals, not negative delay.

The final target class distribution remains imbalanced. In the test set, the On time class represents most observations, while Long delay remains the minority class. Table II shows the final test support used for evaluation.

TABLE II
TEST SET CLASS DISTRIBUTION

Class	Label	Test Support
0	On time (≤ 15 min)	2,453,607
1	Delay (16–90 min)	494,801
2	Long delay (> 90 min)	139,816

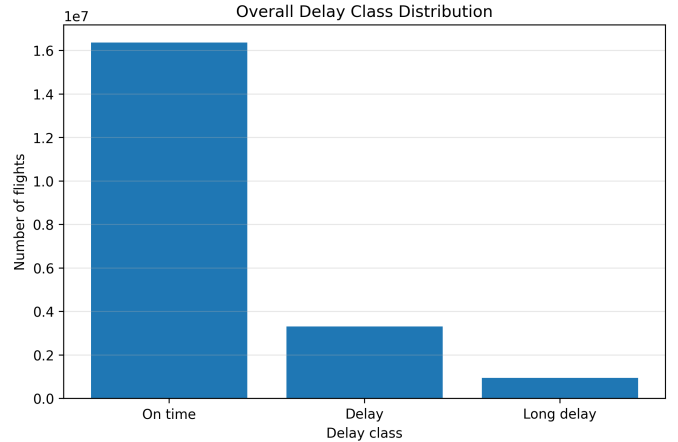


Fig. 1. Overall distribution of the final three-class delay severity target.

B. Target Variable Definition

The target variable was created from arrival delay minutes. Let d_i be the cleaned arrival delay in minutes for flight i , where early arrivals are set to zero. The target y_i is defined as:

$$y_i = \begin{cases} 0, & d_i \leq 15 \\ 1, & 16 \leq d_i \leq 90 \\ 2, & d_i > 90 \end{cases} \quad (1)$$

This target definition balances operational meaning and statistical stability. The first threshold follows the standard 15-minute delay definition used in aviation reporting [2]. The

second threshold separates common manageable delays from longer disruptions.

C. Train, Validation, and Test Split

A stratified random split was used instead of the previous month-based split. The final split proportions were:

- **Train:** 70%
- **Validation:** 15%
- **Test:** 15%

The split was stratified by the target class. This was done to preserve class proportions across all subsets, which is important because the dataset is imbalanced. Since the dataset now includes three complete years of flight records, the stratified random split also helps represent seasonal patterns from different years across training, validation, and test sets.

Table III summarizes the final split structure.

TABLE III
STRATIFIED SPLIT STRUCTURE

Split	Percentage	Records
Train	70%	14,411,712
Validation	15%	3,088,224
Test	15%	3,088,224

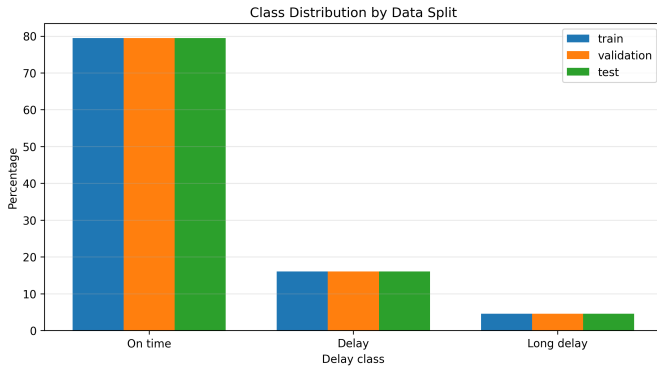


Fig. 2. Class distribution across the stratified train, validation, and test splits.

D. Feature Engineering

The feature engineering stage produced two groups of model inputs: categorical features and numerical features.

The categorical features were:

- operating carrier,
- origin airport,
- origin state,
- destination airport,
- destination state,
- route,
- departure time block,
- arrival time block.

The numerical features were:

- distance,
- scheduled departure minutes of day,

- sine and cosine transformations of scheduled departure time,
- sine and cosine transformations of month,
- sine and cosine transformations of day of week,
- weekend indicator.

Table IV summarizes the feature groups.

TABLE IV
FEATURE GROUPS USED FOR TRAINING

Type	Features
Categorical	Carrier, origin, origin state, destination, destination state, route, departure time block, arrival time block
Numerical	Distance, scheduled departure minutes, cyclic departure time features, cyclic month/day features, weekend indicator

Categorical features were encoded as integer IDs and used with embedding layers. Encoders were fitted only on the training set. Unknown validation or test categories were mapped to zero. Numerical features were scaled using a StandardScaler fitted only on the training data.

Time variables were transformed cyclically. For example, scheduled departure time was converted to minutes of day and then encoded as:

$$x_{\sin} = \sin\left(2\pi \frac{m}{1440}\right), \quad (2)$$

$$x_{\cos} = \cos\left(2\pi \frac{m}{1440}\right), \quad (3)$$

where m is the number of minutes since midnight. This representation avoids treating 23:59 and 00:01 as far apart.

E. Categorical Cardinality and Embeddings

High-cardinality categorical variables are difficult to represent with one-hot encoding. The route feature had 7,454 unique categories, while origin and destination each had 363 categories. Embeddings provide a compact dense representation and allow the model to learn similarities between routes, airports, and carriers.

TABLE V
CATEGORICAL FEATURE CARDINALITIES

Feature	Cardinality
ROUTE	7,454
ORIGIN	363
DEST	363
ORIGIN_STATE_ABR	53
DEST_STATE_ABR	53
DEP_TIME_BLK	20
ARR_TIME_BLK	20
OP_UNIQUE_CARRIER	16

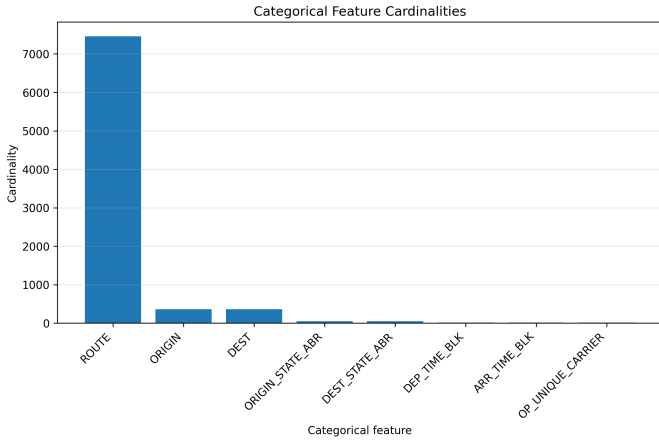


Fig. 3. Cardinality of categorical features used by the embedding-based models.

F. Leakage Prevention

A key methodological decision was to avoid data leakage. The model was designed to predict delay severity using only information that would be available before the flight or from the flight schedule. Therefore, post-flight or delay-cause columns were excluded from the model features.

Excluded columns included actual departure time, departure delay, arrival delay, taxi time, air time, carrier delay, weather delay, NAS delay, late aircraft delay, security delay, cancellation code, and cleaned arrival delay minutes. These variables would make the model appear much stronger, but they are not valid for a pre-flight prediction setting.

G. Class Weights

Class weights were calculated from the training split to reduce the effect of class imbalance. These weights help the model pay more attention to Delay and Long delay examples. However, very aggressive weighting or oversampling was found to destabilize training in early experiments. The final models used controlled and softened class-weight transformations rather than extreme sampling strategies.

IV. MODEL ARCHITECTURES

A. MLP with Embeddings

The first model was an MLP with categorical embeddings. Each categorical variable was passed through an embedding layer. The embeddings were concatenated with 9 scaled numerical features and passed through a feedforward network.

The hidden layers had sizes 256, 128, and 64. Batch normalization, ReLU activations, and dropout of 0.3 were applied. The MLP served as the baseline neural architecture.

The MLP used AdamW optimization, a learning rate of 0.001, weight decay of 10^{-4} , batch size 4096, and CrossEntropyLoss with softened class weights. The model was trained for 10 epochs.

B. Wide and Deep Neural Network

The second model was a Wide and Deep neural network. It had two paths:

- a wide path that mapped numerical features directly to output logits,
- a deep path that processed categorical embeddings and numerical features through dense layers.

The deep path used hidden layers of 384, 192, and 96 units. The final logits were computed as the sum of the wide and deep logits. This architecture was designed to combine simple direct numerical relationships with nonlinear feature interactions.

The Wide and Deep model used AdamW, learning rate 0.001, weight decay 10^{-4} , batch size 4096, and controlled class weights. It was trained for 10 epochs.

C. TabTransformer-Style Model

The third model was a TabTransformer / FT-Transformer style architecture. It represented tabular features as tokens:

- categorical tokens from embedding layers,
- numerical tokens from learned numerical projections,
- one CLS token for global representation.

The token sequence was processed by a Transformer Encoder with:

- $d_{model} = 96$,
- 6 attention heads,
- 4 encoder layers,
- feedforward dimension 384,
- GELU activation,
- transformer dropout 0.12.

The final representation combined the CLS token output with mean token pooling. This representation was concatenated with the numerical features and passed to a classification head with hidden layers of 384 and 192. This pooling strategy was used to avoid relying only on the CLS token and to capture information from the full set of feature tokens.

Training used AdamW [8], OneCycleLR, mixed precision, gradient clipping, label smoothing, and early stopping. Probability calibration was applied after training using the validation set. This calibration multiplied class probabilities by validation-tuned class multipliers and then renormalized the result. The best multipliers were:

$$[1.0, 1.16, 1.305].$$

Calibration was not tuned on the test set.

V. EXPERIMENTAL SETUP

All experiments were run on an HP OMEN 16 laptop with an NVIDIA GeForce RTX 4060 Laptop GPU, an Intel Core i9-13900HX CPU, and 32 GB of DDR5 5600 MHz RAM. The batch size was 4096 for all models.

The training setup is summarized in Table VI. Training time is reported because model complexity and training cost are important in practical deep learning workflows.

TABLE VI
TRAINING CONFIGURATION AND TIME

Model	Epochs	Time (min)
MLP with Embeddings	10	59.90
Wide and Deep	10	54.49
TabTransformer	12	175.52

A. Evaluation Metrics

Accuracy, Weighted F1-score, Macro F1-score, and Balanced Accuracy were considered. Accuracy is useful but can be misleading under imbalance. Weighted F1-score is affected by class support and is therefore also influenced by the majority class. Macro F1-score gives equal weight to each class and was selected as the main metric.

For a class c , precision and recall are:

$$Precision_c = \frac{TP_c}{TP_c + FP_c}, \quad (4)$$

$$Recall_c = \frac{TP_c}{TP_c + FN_c}. \quad (5)$$

The F1-score for class c is:

$$F1_c = 2 \cdot \frac{Precision_c \cdot Recall_c}{Precision_c + Recall_c}. \quad (6)$$

Macro F1 is the average F1 across all classes:

$$MacroF1 = \frac{1}{C} \sum_{c=1}^C F1_c. \quad (7)$$

Because $C = 3$ in this project, each class contributes equally to the final Macro F1-score.

VI. EXPERIMENTAL RESULTS

A. Overall Model Performance

Table VII presents the final test results. The raw TabTransformer achieved the highest accuracy and weighted F1-score. However, the calibrated TabTransformer obtained the best Macro F1-score and balanced accuracy, so it was selected as the final model.

TABLE VII
FINAL TEST PERFORMANCE

Model	Acc.	Bal. Acc.	Macro F1	Weighted F1
MLP with Embeddings	0.7162	–	0.4192	0.7179
Wide and Deep	0.6539	–	0.4149	0.6817
TabTransformer Raw	0.7344	0.4259	0.4309	0.7284
TabTransformer Calibrated	0.7057	0.4425	0.4346	0.7155

The results show an important trade-off. The MLP achieved a strong baseline and obtained better Macro F1 than the Wide and Deep model. The Wide and Deep model predicted delayed classes more often, but this reduced its accuracy and did not improve Macro F1 over the MLP. The raw TabTransformer achieved the best accuracy, while the calibrated TabTransformer achieved the best Macro F1 and balanced accuracy.

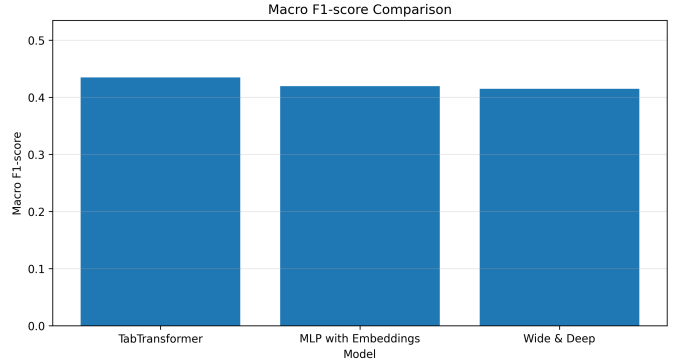


Fig. 4. Macro F1-score comparison of the evaluated models.

B. Per-Class Results

Table VIII shows per-class F1-scores. The On time class is the easiest class for all models. Delay is more difficult, and Long delay is the most difficult class. This is expected because Long delay is the minority class and is likely influenced by external operational conditions not present in the feature set.

TABLE VIII
PER-CLASS F1-SCORE ON THE TEST SET

Model	On time	Delay	Long delay
MLP	–	–	–
Wide and Deep	0.78	0.34	0.12
TabTransformer Raw	0.85	0.30	0.15
TabTransformer Calibrated	0.83	0.31	0.17

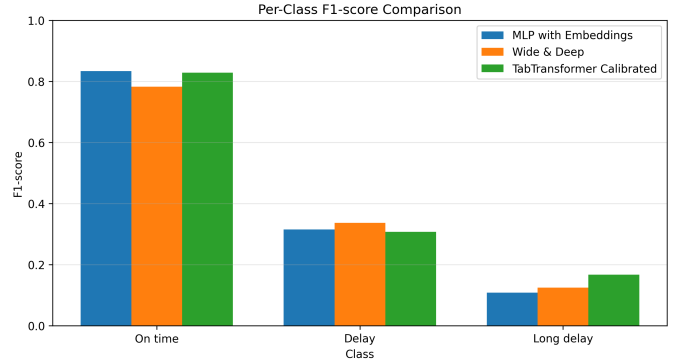


Fig. 5. Per-class F1-score comparison across the main models.

Table IX shows the per-class recall values for the main reported models. The calibrated TabTransformer increases Long delay recall compared with the raw TabTransformer, improving from 0.13 to 0.18. This improvement is important because Long delay is the most difficult and operationally relevant minority class.

C. Confusion Matrix Analysis

The confusion matrix analysis shows that Long delay remains the most difficult class. Many Long delay flights are

TABLE IX
PER-CLASS RECALL ON THE TEST SET

Model	On time	Delay	Long delay
Wide and Deep	0.72	0.49	0.10
TabTransformer Raw	0.86	0.29	0.13
TabTransformer Calibrated	0.81	0.33	0.18

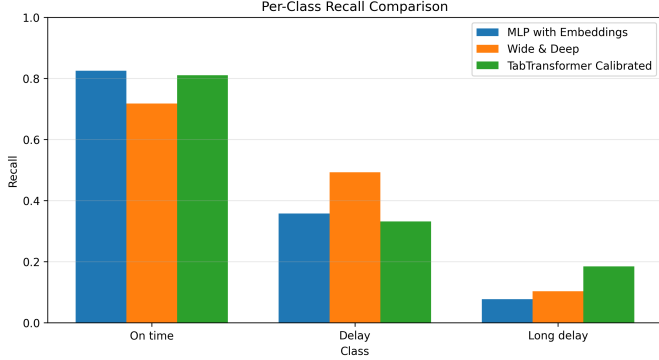


Fig. 6. Per-class recall comparison across the main models.

still classified as On time or Delay. This is a limitation of the current feature set. However, the calibrated TabTransformer predicts more Delay and Long delay cases than the raw version, which improves Macro F1 and balanced accuracy.

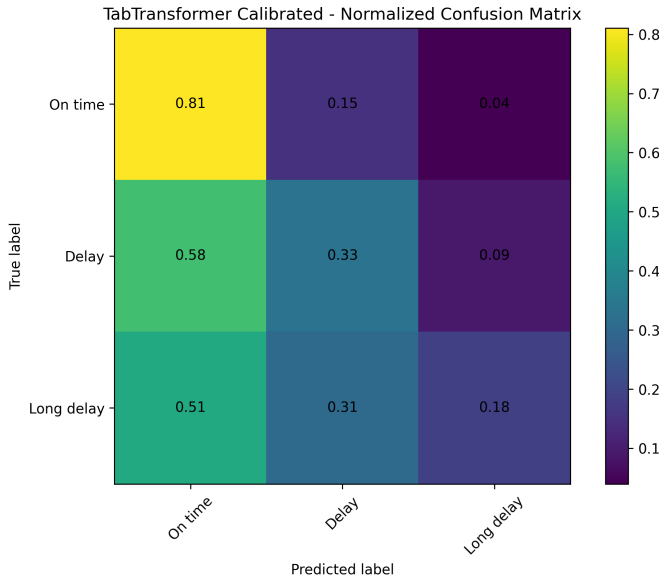


Fig. 7. Normalized confusion matrix for the calibrated TabTransformer.

D. Training Behavior

The TabTransformer trained for 12 epochs and achieved its best raw validation Macro F1-score of 0.4314. Training was more expensive than the MLP and Wide and Deep models because the Transformer Encoder processes categorical and numerical feature tokens with self-attention.

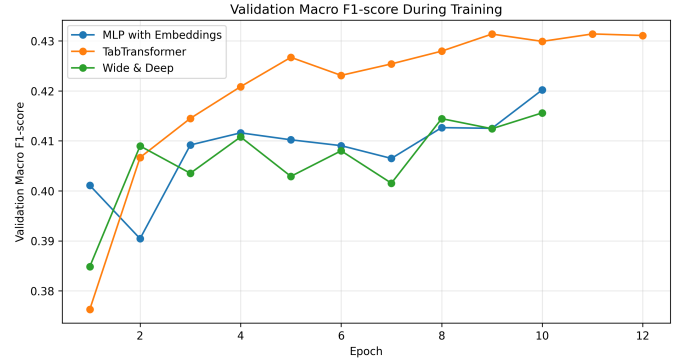


Fig. 8. Validation Macro F1-score during training.

VII. GRAPHICAL INFERENCE PROTOTYPE

A small Python graphical application was implemented to demonstrate how the trained model can be used for inference. The application loads the calibrated TabTransformer checkpoint, metadata, categorical encoders, numerical scaler, and route lookup file. The user selects flight information from dropdown lists, including carrier, route, month, day of week, scheduled departure time, and arrival time block.

The interface displays calibrated probabilities for the three delay severity classes. This makes the model easier to interpret than only returning a hard class prediction. The prototype is not a production system, but it shows how the trained model can be converted into a practical decision-support tool.

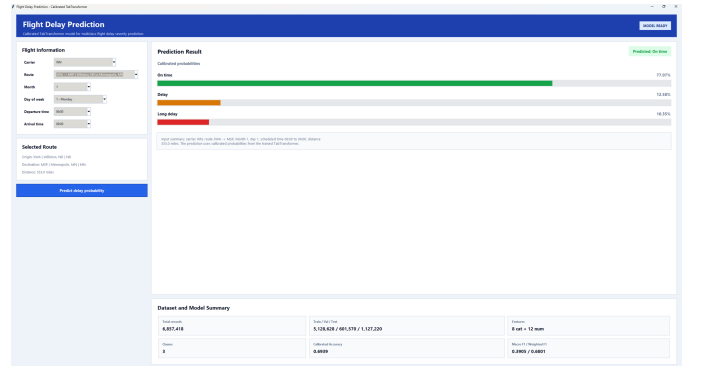


Fig. 9. Graphical inference interface developed in Python to estimate calibrated delay severity probabilities using the trained TabTransformer model.

VIII. DISCUSSION

The experimental results should be interpreted according to the difficulty of the task. The model only uses information available before the flight or from the schedule, such as airline, route, origin, destination, scheduled departure time, arrival time block, distance, and calendar features. It does not use weather, aircraft rotation, previous flight delay, maintenance events, airport congestion, air traffic control restrictions, or real-time operational disruptions.

These missing factors are likely important for predicting Long delay. A long delay can be caused by unexpected

events that are not visible from schedule information alone. Therefore, the relatively low Long delay recall is expected. The model should be interpreted as an early risk estimator rather than a definitive operational predictor.

The stratified random split helped preserve class proportions and seasonal variability across training, validation, and test sets. This was useful after extending the dataset to three years, because each subset includes examples from different seasons and operational periods.

The model comparison shows that accuracy alone is not enough. The raw TabTransformer achieved the best accuracy and weighted F1-score, but the calibrated TabTransformer achieved better Macro F1 and balanced accuracy. This means that calibration reduced the dominance of the majority class and improved minority-class behavior. In an imbalanced task, this is more important than maximizing accuracy only.

The improvement over the MLP and Wide and Deep models is moderate, not dramatic. However, it is meaningful because it occurs in the most difficult part of the problem: minority-class delay severity detection. In real operational settings, missing a Long delay can be more costly than misclassifying an On time flight as a moderate Delay risk. Therefore, selecting the best model by Macro F1 is more appropriate than selecting only by accuracy.

IX. CONCLUSION AND FUTURE WORK

This paper presented a complete deep learning pipeline for flight delay severity prediction using 2023–2025 flight records. The final model-ready dataset contained 20,588,160 flights. A three-class target was used to classify flights as On time, Delay, or Long delay. This target definition improved stability compared with the original five-class design, which produced severe imbalance.

Three PyTorch models were evaluated: an MLP with embeddings, a Wide and Deep neural network, and a TabTransformer-style architecture. The MLP provided a strong baseline with a Macro F1-score of 0.4192. The Wide and Deep model predicted delayed classes more often, but did not outperform the MLP in Macro F1. The raw TabTransformer achieved the best accuracy, while the calibrated TabTransformer achieved the best Macro F1-score of 0.4346 and the best balanced accuracy of 0.4425. Therefore, the calibrated TabTransformer was selected as the final model.

The results are meaningful because the evaluation avoids direct data leakage and uses only schedule-based or pre-flight information. However, Long delay prediction remains difficult because the model does not use external operational features. Future work should integrate weather data, aircraft rotation information, previous flight delay, airport congestion indicators, maintenance events, and real-time air traffic conditions. Additional experiments could also compare deep learning models against gradient boosting baselines, evaluate repeated runs with confidence intervals, and explore cost-sensitive thresholds for operational decision-making.

Finally, the graphical inference prototype could be extended into a web dashboard or operational monitoring tool. This

would make the model easier to test with different flight scenarios and could support practical delay risk estimation.

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